

Vidhi Joshi

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EDUCATION

Western University

Sep 2022 - Aug 2026

BSc. Honours, Computer Science (GPA: 3.2)

London, Ontario

- **Coursework:** Systems Programming, Operating Systems, Data Structures & Algorithms, Computer Graphics, Intro to AI, Object-Oriented Design & Analysis, Databases, Game Design

EXPERIENCE

Haudenosaunee POD | *Research assistant*

Mar 2025 - Present

- Architected a production-grade digital archival system using DSpace and Docker, containerizing the lab's infrastructure to ensure 99.9% uptime for Indigenous AI training datasets.
- Engineered an automated OCR and OpenCV pipeline to digitize and clean 1,000+ PDF scans, creating a high-fidelity data repository for LLM training.
- Designed embedded system dashboards using Arduino and Raspberry Pi to monitor real-time lab workflows, increasing researcher efficiency by 30%.
- Developed 3D assets and media visualizations using OpenGL and Blender to support interactive research presentations.

Riipen - Owning My | *Growth Project Lead*

Mar 2026 - Apr 2026

- Developed a scalable product growth strategy to identify and implement new market opportunities through targeted engineering workflows.
- Orchestrated a cross-functional team to merge technical feasibility with product development, delivering a unified implementation roadmap.
- Designed predictive intent scoring and automation strategies to refine technical decision-making and product differentiation.

Softsages Technologies | *Tech Intern*

May 2023 - Aug 2023

- Engineered production-grade chatbots for Telegram and WhatsApp using custom APIs to implement an action layer for conversational UI flows, boosting user engagement by 40%.
- Built and deployed client-facing web modules using iterative prototyping to ensure seamless integration between frontend UI and backend logic, reducing response errors by 25%.
- Utilized Postman to create APIs and webhooks for chatbots to efficiently store interaction data in CSV files.

PROJECTS

Artispace | *Personal Project*

Nov 2025 - Apr 2026

- Conceptualized and architected a high-fidelity social portfolio platform with a custom many-to-many database schema for exhibition curation.
- Designed a dynamic UI rendering engine allowing layout customization via artist-defined JSON configuration files and Masonry/Grid views.
- Optimized media delivery by implementing an affinity-based ranking algorithm and an asynchronous image processing pipeline to maintain original fidelity.

SKILLS

- **Core Engineering:** Python, C++, Java, SQL, JavaScript (Node.js/React), REST APIs, Git/GitHub
- **AI & Computer Vision:** OpenCV, OCR, RAG (Retrieval-Augmented Generation), Prompt Engineering
- **Graphics & High-Performance:** OpenGL, OpenCL, GLSL, Shaders, Parallel Computing
- **Cloud & DevOps:** Docker, Docker Compose, Linux Server Admin (Ubuntu), DSpace, Bash Scripting, CI/CD Basics
- **Design & Creative:** Figma, Adobe Creative Suite, 3D Modeling, UI/UX Prototyping, Wireframing, User Research
- **Hardware & Embedded:** Arduino, Raspberry Pi, Embedded Systems, IoT Prototyping
- **Technical Foundations:** Data Structures & Algorithms, Object-Oriented Design (OOD), UML, Discrete Math, Computer Logic, Databases